

**Draft Proposal
for
Design, Development, and Maintenance
Of
Senior Citizen Portal (Kutumb Portal)
Delhi Police**



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1. Executive Summary

1.1. Project Overview

This proposal outlines the development of a comprehensive digital ecosystem for the Delhi Police Senior Citizen Portal. The project aims to modernize and streamline the interaction between the Delhi Police and the senior citizen community by creating a suite of integrated digital tools. The solution comprises two distinct mobile applications (one for citizens, one for police officers) available on both Android and iOS, a powerful web-based Admin Console for operational management, and a public-facing web portal for citizen registration. This integrated platform will enhance both service delivery for citizens and operational effectiveness for the police force.

1.2. Problem Statement & Proposed Solution

Problem Statement: The current system for managing senior citizen welfare and safety faces several challenges. Verification visits are often tracked manually, leading to inefficiencies, a lack of real-time oversight, and difficulty in ensuring accountability. Senior citizens lack a dedicated, user-friendly digital channel to request services, report emergencies instantly, or manage their information with the police. This operational gap can lead to delayed responses, incomplete data, and a reactive rather than proactive approach to senior citizen safety.

Proposed Solution: We propose a holistic digital solution built around a central backend infrastructure. The solution will:

1. **Empower Senior Citizens:** A user-friendly mobile application will allow citizens to register, manage their profiles, access essential e-services (e.g., tenant verification, theft e-FIR), and, most critically, trigger an SOS alert to family and police with their live location during emergencies.
2. **Equip Police Officers:** A dedicated officer mobile application will automate the visit and assessment process. Officers can view assigned visits, locate nearby citizens on a map, and conduct on-site verifications. Accountability is enforced through mandatory GPS-based check-ins (within a 30-meter geo-fence) and live photo capture with "liveness" detection to validate the visit.
3. **Enable Administrative Control:** A comprehensive Admin Console will provide a centralized dashboard for managing officer rosters, assigning visits, approving reports, and generating analytics, giving supervising officers complete visibility and control over the program.

1.3. Key Objectives

The primary objectives of this project are to:

- **Enhance Citizen Safety:** Provide an immediate and reliable SOS channel for senior citizens in distress.

- **Improve Service Accessibility:** Offer a single point of access for various police e-services, making them more convenient for senior citizens.
- **Increase Operational Efficiency:** Automate the assignment, tracking, and reporting of officer visits, reducing administrative overhead.
- **Ensure Accountability & Transparency:** Implement a foolproof verification system using GPS, live photos, and liveness checks to ensure visits are genuinely conducted.
- **Centralize Data Management:** Create a secure, unified database of all registered senior citizens, their visit histories, and risk assessments.
- **Facilitate Data-Driven Policing:** Provide analytics and reports to help the Delhi Police identify trends, allocate resources effectively, and proactively address the needs of the senior citizen community.

1.4. Scope of Work (Phase 1)

The scope for the initial phase of this project includes the design, development, security audit, deployment, and support of the following core deliverables:

- 1. Citizen Mobile App (Android & iOS):**
 - Mobile/OTP based registration and login.
 - Profile management (personal details, contacts, health risk).
 - Access to five key e-services via WebView.
 - Emergency SOS button with location sharing.
 - Timeline view for "My Requests & Visit History."
 - In-app notification system.
- 2. Officer Mobile App (Android & iOS):**
 - Secure login using Official ID and OTP.
 - Dashboard with key metrics and visit suggestions.
 - Map-based view of senior citizens within the officer's beat.
 - Complete visit management module (Assigned, Completed, All).
 - Geo-fenced (30m) visit check-in and assessment questionnaire.
 - Live photo and liveness verification system.
- 3. Admin & Sr. Citizen Web Consoles:**
 - **Admin Console:** For officer management, roster assignment, visit approval workflows, citizen profile management, and report generation.
 - **Sr. Citizen Console:** A public web portal for self-registration and profile editing.
- 4. Backend & Infrastructure:**
 - Development of all necessary backend APIs for authentication, workflows, and data management.
 - Data migration from the existing Senior Citizen Portal.
 - Deployment on secure, government-approved cloud infrastructure (NIC / MeghRaj).
- 5. Compliance & Support:**
 - Mandatory VAPT and security compliance audit by a CERT-In impanelled vendor.
 - A 6-month post-Go-Live maintenance and support period for all components.

2. Proposed Solution Architecture

This section details the technical architecture of the proposed Delhi Police Senior Citizen Portal ecosystem. The architecture is designed to be secure, scalable, and robust, ensuring reliable service delivery and efficient operations.

2.1. System Components Overview

The solution is composed of several interconnected components, each serving a specific function within the ecosystem:

1. **Citizen Mobile Application (Android & iOS):** The primary interface for senior citizens. It provides features for registration, profile management, service requests, and a critical SOS button for emergencies. It communicates exclusively with the central backend via secure REST APIs.
2. **Officer Mobile Application (Android & iOS):** The field tool for police officers. It facilitates secure login, visit management, real-time location tracking of nearby citizens, and a structured workflow for conducting and submitting on-site assessments with geo-fencing and liveness verification.
3. **Admin Web Console:** A comprehensive web-based portal for supervisory staff (SHO/DO) and administrators. It serves as the command center for managing officer rosters, assigning and approving visits, viewing citizen data, and generating analytical reports.
4. **Senior Citizen Web Console:** A public-facing web portal that offers an alternative to the mobile app, allowing citizens to register and manage their profiles from a desktop or laptop computer.
5. **Central Backend (API Server):** The core of the system. It is built on a service-oriented architecture and exposes a set of secure REST APIs. It handles all business logic, including user authentication, data processing, workflow management, and integration with third-party services.
6. **Database:** A centralized and secure database (e.g., PostgreSQL, MySQL) hosted on government-approved cloud infrastructure. It will store all application data, including user profiles, visit records, assessment data, and system logs.
7. **Third-Party Services:** The platform integrates with essential external services to enhance functionality:
 - **Google Maps API:** For mapping, location services, directions, and geo-fencing calculations.
 - **SMS/OTP Gateway:** For sending one-time passwords for secure user authentication.
 - **Firebase Cloud Messaging (FCM):** For delivering real-time push notifications to both citizen and officer mobile applications.

2.2. High-Level Architecture Diagram

The system follows a modern, multi-tiered architecture that separates the presentation layer (client applications) from the business logic (backend) and data storage. This ensures modularity, security, and scalability.

The architecture consists of the following layers:

- **Presentation Layer (Clients):** This includes the native mobile applications (Android/iOS for Citizens and Officers) and the web-based consoles (Admin and Citizen). All clients communicate with the backend via secure HTTPS requests.
- **Application Layer (Backend):** This is the central API server that processes all incoming requests, enforces business rules, and coordinates with the data and services layers. It acts as the single source of truth for the entire ecosystem.
- **Data Layer:** This layer consists of the primary database that securely stores all application data.
- **Services Layer:** This includes all external third-party integrations, such as Google Maps, FCM, and OTP services, which are accessed by the backend server to provide specific functionalities.

2.3. User Workflow & Interaction Model

The interaction between components is best understood through the primary user workflows:

Workflow 1: Citizen Registration and Verification

1. A new senior citizen registers through the Citizen Mobile App or Web Console.
2. The client application sends the registration data to the Backend API.
3. The Backend creates a citizen profile with a "Pending Verification" status and generates a new visit task.
4. An administrator in the **Admin Console** views the pending task and assigns it to a beat officer.
5. The Backend sends a push notification via **FCM** to the assigned officer's mobile app.
6. The officer conducts the visit, completes the assessment on the **Officer App**, and submits the report.
7. The Backend updates the citizen's profile to "Verified" and notifies the citizen via a push notification.

Workflow 2: Officer Visit and Assessment

1. An officer logs into the **Officer App** using their Official ID and an **OTP**.
2. The officer views their assigned visits or uses the "Nearby Citizens" map, powered by **Google Maps API**, to locate a citizen.
3. Upon arrival, the app verifies the officer is within the 30-meter geo-fence.
4. The officer initiates the assessment, fills out the questionnaire, and captures a live photo.
5. The completed assessment, including the photo, timestamp, and GPS coordinates, is sent to the Backend.
6. The visit report appears in the **Admin Console** for a supervising officer (SHO/DO etc) to review and approve.
7. Once the application is approved and the address is verified, Citizens will get a Card with basic Details.
8. In Routine visits, the officer will do risk assessment and will submit the updated report. All details will be logged with version control with time and data stamp.

Workflow 3: Citizen Emergency (SOS)

1. A citizen in distress presses the SOS button on their **Citizen App**.
2. The app immediately sends an emergency alert to the Backend, including the citizen's live location, device details, and battery status.
3. The Backend instantly triggers multiple actions:
 - Sends SMS alerts via the **OTP/SMS Gateway** to the citizen's registered emergency contacts.
 - Sends a high-priority push notification via **FCM** to the local police station's dashboard in the **Admin Console**.
 - Logs the event and tracks the citizen's location in real-time.

3. Detailed Functional Specifications

This section provides a detailed breakdown of the features and functionalities for each component of the proposed system.

3.1. Citizen Mobile Application (Android & iOS)

This application is the primary touchpoint for senior citizens, designed for ease of use, accessibility, and quick access to emergency services.

3.1.1. Authentication: Registration & Login (Mobile + OTP)

- **Login:** Existing users will log in using their 10-digit mobile number. A time-sensitive One-Time Password (OTP) will be sent via SMS to the registered number for secure verification.

- **Registration:** First-time users will register by providing their mobile number and Date of Birth. The system will validate the mobile number via OTP. Upon successful registration, the user's device details (IMEI/Device ID) will be captured and linked to the profile for enhanced security.

3.1.2. Profile Verification Workflow

- Upon initial registration, a citizen's profile is created with a "Pending Verification" status.
- This action automatically generates a new verification visit task in the Admin Console.
- The task is assigned to a beat officer who then conducts an in-person visit to validate the citizen's details.
- Only after the officer submits a successful verification report via the Officer App will the citizen's profile be marked as "Verified" and fully activated.

3.1.3. Home Dashboard

- The main screen will display a "Citizen Card" with the user's name, photo, QR Code, and current verification status.
- It will show details of the assigned Police Station and Beat Officer.
- Prominent, one-tap call buttons will be available for connecting to the Emergency Helpline (112).

3.1.4. Services Module (WebView Integration)

- This section will provide a consolidated portal to access key Delhi Police e-services.
- To ensure consistency and rapid integration, these services will be loaded within a secure in-app WebView. The five services for Phase 1 are:
 1. Lost/Find Report
 2. Theft e-FIR
 3. Domestic Help Verification
 4. Tenant Verification

3.1.5. Emergency / SOS System

- A large, clearly marked SOS button will be accessible from the main dashboard.
- When pressed, the system will instantly send an alert containing the citizen's GPS location, profile details, device ID, and their pre-configured emergency contacts (via SMS) and the police Admin Console (via push notification).

3.1.6. Profile Management

- Users can view and edit their personal information, including address, health and emergency contacts.
- **Crucially, any modification to core profile details (e.g., address) will automatically trigger a new verification visit request to ensure data integrity.**

3.1.7. My Requests & Visit History

- This section will provide a chronological timeline of all interactions with the police through the app. It will list all service requests, officer visits (both pending and completed), and their outcomes.

3.1.8. Notification Center

- An inbox for all push notifications sent to the user. This will include alerts about upcoming visits, confirmation of completed visits, and general announcements from the police department.

3.2. Officer Mobile Application (Android & iOS)

This is a secure, internal tool designed to streamline the field operations of police officers.

3.2.1. Secure Authentication (Official ID + OTP)

- Officers will log in using their unique Official ID.
- An OTP will be sent to the officer's mobile number registered in the Admin Console to complete the login, ensuring only authorized personnel gain access. Self-registration is disabled.

3.2.2. Officer Dashboard

- The landing screen will provide a summary of key metrics, such as the number of assigned vs. completed visits.
- It will feature a "Visit Suggestions" panel that prioritizes citizens based on risk level and pending visit duration.

3.2.3. Nearby Citizens Module (Map & List Views)

- **Map View:** An interactive map displaying pins for all senior citizens within the officer's beat, allowing for efficient route planning.
- **List View:** A scrollable list of citizens with searchable and sortable cards. Each card will display the citizen's photo, name, address, risk status, and provide one-tap options to call or get directions via Google Maps.

3.2.4. Visit Management Module (Assigned, Completed, All)

- This module will have three tabs to organize all tasks:
 - **Assigned:** A list of all pending visits.
 - **Completed:** A historical log of all successfully conducted visits with their assessment data.
 - **All Visits:** A comprehensive, searchable history of every visit.

3.2.5. Assessment Workflow (Check-in, Geo-fencing, Photo upload Verification)

- **Geo-fenced Check-in:** The "Start Assessment" button for a citizen will only become active when the officer's device GPS is within a 30-meter radius of the citizen's registered address.
- **Questionnaire:** The officer will fill out a standardized digital form covering health, safety, and vulnerability checks.
- **Submission:** The completed report, along with the geo-tagged photo, timestamp, and officer's GPS coordinates, will be submitted to the backend.

3.2.6. Officer Profile Management

- Officers can view their official details as registered in the system. They can request edits to certain fields, which will be sent to an administrator for review and approval before changes are reflected.

3.3. Admin Console (Web Portal)

The web-based command center for managing the entire senior citizen program.

3.3.1. User Management (Police Officers)

- Administrators can create, view, edit, and deactivate profiles for all police officers using the mobile app.

3.3.2. Roster Management & Visit Assignment

- View all pending visit requests in a centralized queue.
- Assign visits to specific officers either manually or automatically based on beat-level proximity and workload.

3.3.3. SHO/DO Approval Workflow

- A dedicated screen for supervising officers (SHO/DO) to review all submitted visit reports. They can view the assessment details, the liveness-verified photo, and the GPS data before marking the visit as "Approved" or "Reopened" for re-verification.

3.3.4. Senior Citizen Profile Management

- Provides a comprehensive view of all registered senior citizens. Admins can search, filter, and view detailed profiles, including their visit history and risk status.

3.3.5. Reporting & Analytics

- A powerful reporting module to generate operational reports. Data can be filtered by Date Range, District, Police Station, and Beat to analyze visit completion rates, risk distribution, and other key performance indicators.

3.3.6. Geo-Fencing Configuration

- A settings module to manage the parameters for the geo-fencing feature, such as the default radius for visit check-ins.

3.4. Senior Citizens Web Console

A public-facing portal providing web-based access for citizens.

3.4.1. Public User Registration

- Allows senior citizens who may not have a smartphone or prefer using a computer to register for the service through a simple web form. The verification workflow remains the same as for mobile registration.

3.4.2. Web-based Profile Management

- Registered citizens can log in to view and edit their profile information, check their visit history, and manage their emergency contacts, mirroring the functionality of the mobile application's profile section.

4. Technical Architecture & Specifications

This section outlines the technical foundation, technology stack, and integration strategies for the project.

4.1. Technology Stack

- **Mobile Applications (iOS & Android):** Native development will be prioritized to ensure optimal performance, security, and device integration.
 - **iOS:** Swift Native/flutter/reactJS.
 - **Android:** Kotlin Native/flutter/reactJS.
- **Backend:** A robust and scalable server-side framework.
 - **Language/Framework:** Node.js (with Express.js) or. Net Core.
 - **Server Environment:** To be deployed on government-approved cloud infrastructure (NIC / MeghRaj Cloud).
- **Frontend (Admin & Citizen Consoles):** A modern JavaScript framework for a responsive and dynamic user experience.
 - **Framework:** React.js or Angular.
- **Database:** A reliable relational database management system.
 - **System:** PostgreSQL or MySQL.

4.2. Backend & API Architecture

4.2.1. Monolithic Approach

This approach provides a strong, unified codebase that is easier to develop, test, and deploy initially. The application will be internally structured into logical modules (e.g., Authentication, User Management, Visit Management) to ensure separation of concerns and facilitate future scalability.

4.2.2. REST API Specifications

All communication between the client applications and the backend will be conducted over HTTPS using a well-defined RESTful API. Data will be exchanged in JSON format.

4.2.3. Authentication & Authorization (RBAC)

The system will implement robust Role-Based Access Control (RBAC) using JSON Web Tokens (JWT). Upon successful login, the API will issue a JWT containing the user's role. This token will be sent with every subsequent request to secure API endpoints. The defined roles are:

- **Citizen:** Access to their own profile, service requests, and SOS.
- **Officer:** Access to assigned citizen data, visit management, and their own profile.
- **SHO/DO/ACP/DCP/CP (Supervisor):** Access to reports submitted by officers within their jurisdiction.
- **Admin:** Full access to the system for user management, assignments, and reporting.

4.3. Database Design & Schema

The database schema will be designed to be relational and normalized. Key tables will include:

- **Users:** Stores common login information for all user types (citizens, officers, admins).
- **Citizen_Profiles:** Contains detailed information specific to senior citizens.
- **Officer_Profiles:** Contains details specific to police officers, including their beat and station.
- **Visits:** Manages all visit tasks, including status, assignment details, and scheduling.
- **Visit_Assessments:** Stores the data from the questionnaire and the verification photo for each completed visit.
- **SOS_Alerts:** Logs all emergency requests with associated location data.

4.4. Third-Party Integrations

4.4.1. SMS/OTP Gateway Integration

The backend will integrate with a designated SMS gateway provider (e.g., MSG91, Twilio). The API will make a server-to-server call to the provider's endpoint to send OTPs for user authentication.

4.4.2. Google Maps API for Geo-services

- **Client-Side:** The officer and citizen mobile apps will use the Google Maps SDK to render maps and display location data.
- **Server-Side:** The backend will utilize Google Maps APIs for geocoding addresses and calculating distances for the geo-fencing check-in and for automated visit assignment logic.

4.4.3. Firebase Cloud Messaging (FCM) for Push Notifications

The backend will integrate with FCM to send real-time notifications. Each mobile app instance will register with FCM to receive a unique device token, which will be stored in the database. The backend will use these tokens to push targeted alerts, such as new visit assignments for officers or visit status updates for citizens.

4.5. Data Migration Strategy

A phased approach will be taken to migrate data from the existing Delhi Police Senior Citizen Portal to the new system.

1. **Extraction:** Develop scripts to extract all relevant data from the current database into a standardized format (e.g., CSV).
2. **Transformation:** Clean, validate, and transform the extracted data to match the new database schema.
3. **Loading:** Import the transformed data into the new database.
4. **Validation:** Perform rigorous checks and run parallel tests to ensure data integrity and completeness before the final switchover.

5. Infrastructure & Deployment

This section covers the hosting environment, deployment methodologies, and scalability planning for the project.

5.1. Hosting Environment (NIC / MeghRaj Cloud)

The entire backend infrastructure, including the API server and database, will be hosted on a secure, government-approved cloud platform such as NIC Cloud or MeghRaj Cloud. This ensures compliance with government data residency and security policies. The environment will be configured with separate instances for production, staging, and development to ensure a stable and controlled release process.

5.2. Deployment Strategy (CI/CD)

A Continuous Integration and Continuous Deployment (CI/CD) pipeline will be established to automate the build, testing, and deployment processes.

- **Version Control:** All code will be managed in a Git repository.
- **CI/CD Pipeline:** A tool like Jenkins or GitLab CI will be used to automatically trigger builds upon code commits. The pipeline will include steps for static code analysis, unit testing, and integration testing.
- **Deployment:** Successful builds will be automatically deployed to the staging environment for UAT. Deployments to the production environment will be managed through a controlled, approval-based process.

5.3. Scalability & Performance Considerations

The architecture is designed for scalability to handle a growing user base.

- **Load Balancing:** A load balancer will be configured to distribute incoming traffic across multiple instances of the API server, ensuring high availability and preventing any single point of failure.
- **Stateless Backend:** The API server will be designed to be stateless, allowing for horizontal scaling by simply adding more server instances as needed.
- **Database Optimization:** The database will be optimized with proper indexing and query analysis to ensure fast response times. Read replicas can be added in the future to handle increased read loads.

5.4. Mobile Application Deployment (Google Play Store & Apple App Store)

- **Submission:** The native Android and iOS applications will be packaged and submitted to the Google Play Store and Apple App Store, respectively.
- **Review Process:** The submission will adhere to all platform-specific guidelines and will undergo the mandatory review process of each store.
- **Release Management:** A phased rollout (e.g., for a small percentage of users initially) will be considered to monitor for any unforeseen issues before a full release. App updates and new versions will be managed through the same deployment pipeline.

6. Security, Compliance & Data Privacy

This section details the measures that will be taken to ensure the security, integrity, and privacy of all user data.

6.1. Data Encryption (In-Transit & At-Rest)

- **In-Transit:** All communication between the mobile applications, web consoles, and the backend API server will be encrypted using industry-standard TLS 1.3/1.2 protocols (HTTPS). This prevents eavesdropping and man-in-the-middle attacks.
- **At-Rest:** All sensitive user data stored in the database on the NIC / MeghRaj Cloud infrastructure will be encrypted at the storage level. This protects data from unauthorized access even in the event of a physical security breach.

6.2. Role-Based Access Control (RBAC)

The system is built on a strict RBAC foundation, as detailed in Section 4.2.3. Access to data and system functionalities is strictly limited based on the user's role (Citizen, Officer, SHO/DO, Admin). This ensures that users can only view and interact with the information and tools necessary for their specific duties, adhering to the principle of least privilege.

6.3. CERT-In Security Audit & VAPT Compliance

Prior to the Go-Live date, the entire system—including the mobile applications, web portals, and backend APIs—will undergo a mandatory and comprehensive security audit. This Vulnerability Assessment and Penetration Testing (VAPT) will be conducted by a CERT-In empanelled vendor. All identified vulnerabilities will be remediated to ensure the platform is secure and compliant with government security standards before it is launched for public and official use.

7. Project Plan & Timeline

This section outlines the proposed project schedule, key milestones, and the testing plan to ensure a successful and timely delivery.

7.1. Project Phases & Milestones

The project will be executed in a phased manner to ensure structured development and quality control.

Phase	Window	Milestone
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Phase 1: Discovery & Planning	Weeks 1-2	Finalization of detailed requirements, technical architecture, and UI/UX wireframes. Project kickoff meeting.
Phase 2: Core Backend & Admin Console Development	Weeks 3-8	Backend APIs and database schema complete. Admin Console with user and visit management functionalities is functional.
Phase 3: Mobile Application Development	Weeks 7-12	Citizen and Officer mobile applications (iOS & Android) are feature-complete and ready for integration.
Phase 4: Integration & Internal Testing	Weeks 11-14	All components are successfully integrated. The system passes all internal QA tests, including unit, integration, and system testing.
Phase 5: UAT, Security Audit & Deployment	Weeks 15-16	Successful completion of UAT with stakeholder sign-off. VAPT report from CERT-In vendor received and vulnerabilities remediated. Final applications deployed to app stores and production servers.
Phase 6: Post-Launch Support & Maintenance		Stable system operation with ongoing support for bug fixes, performance monitoring, and necessary updates.

7.2. User Acceptance Testing (UAT) Plan

The UAT is the final validation stage to ensure the solution meets the operational needs of the Delhi Police and its senior citizen users.

- **Objective:** To receive formal confirmation from key stakeholders that the system functions as per the agreed-upon requirements and is ready for production deployment.
- **Participants:** A designated group of end-users, including Delhi Police officers (from Constable to SHO levels) and a sample group of senior citizen volunteers.
- **Process:**

1. **Test Case Preparation:** Detailed test scripts will be prepared covering all user workflows, from a citizen registering and using the SOS feature to an officer conducting and submitting a visit assessment.
2. **Staging Environment:** UAT will be performed on a dedicated staging server that is a replica of the production environment, ensuring realistic testing conditions.
3. **Execution & Feedback:** Participants will execute the test scripts. A formal process, using a shared issue tracker, will be established for participants to report any bugs, provide feedback, or request clarifications.
4. **Review & Resolution:** Daily review meetings will be held to discuss feedback and prioritize the resolution of any identified issues.
5. **Sign-off:** The UAT phase concludes once all critical and major issues have been resolved to the satisfaction of the stakeholders, who will then provide a formal sign-off for the system's Go-Live.

8. Project Deliverables (Phase 1)

Upon successful completion of the project, the following items will be delivered.

8.1. Source Code for all Components

The complete, well-documented source code for all developed components will be delivered. This includes the iOS application, Android application, Backend API, Admin Console, and the Senior Citizens Web Console. The code will be provided via a secure, private Git repository.

8.2. Deployed Mobile Applications (Android & iOS)

The final, compiled, signed, and production-ready mobile applications will be submitted and published on the Google Play Store and the Apple App Store, making them publicly available for download by senior citizens and for internal distribution to officers.

8.3. Deployed Web Consoles (Admin & Citizen)

The fully functional and deployed Admin Console and Senior Citizens Web Console will be hosted on the designated NIC / MeghRaj Cloud infrastructure, accessible via their respective secure URLs.

8.4. Technical Documentation

Comprehensive technical documentation will be provided, including:

- **API Documentation:** A detailed guide for all API endpoints (e.g., Swagger or OpenAPI specification).
- **Database Schema:** A visual and descriptive representation of the database structure.
- **Architecture Diagrams:** High-level diagrams illustrating the system architecture and user flows.
- **Deployment & Maintenance Guide:** Instructions for deploying, configuring, and maintaining the system.

8.5. Security Audit Compliance Report

The final, official report from the CERT-In empanelled VAPT vendor. This document will certify that a thorough security audit has been conducted and that all identified vulnerabilities have been successfully remediated, confirming the platform's compliance with government security standards.

9. Maintenance & Support

This section details the plan for ongoing support and maintenance after the system goes live to ensure its long-term stability and performance.

9.1. Post-Go-Live Support Plan (6 & 12 Months)

A comprehensive support plan will be provided to ensure the smooth operation of the entire platform.

- **Standard Support (6 Months):** A 6-month support period is included for all components (Backend, Web Consoles, Mobile Apps) as standard.
- **Extended Support (12 Months):** An optional 12-month extended support plan is available specifically for the iOS and Android mobile applications.
- **Services Included:**
 - **Technical Assistance:** A dedicated point of contact for resolving technical queries.
 - **Bug Fixing:** Prioritized resolution of any software defects or errors.
 - **Performance Monitoring:** Continuous monitoring of server and application performance to identify and address bottlenecks.
 - **Version Upgrades:** Management of application versioning and necessary updates.

9.2. Service Level Agreement (SLA)

A clear SLA will be in place to govern the response and resolution times for any reported issues, categorized by severity.

Severity Level	Description	Response Time	Resolution Target
Critical	System-wide outage; core functionalities (e.g., SOS, Login) are unavailable.	2-4 Business Hours	24 Hours
Major	A significant feature is non-functional with no viable workaround.	8 Business Hours	3-5 Business Days
Minor	A minor feature issue, UI defect, or inconvenience with a workaround available.	24 Business Hours	10 Business Days

9.3. Bug Fixing & OS Upgrade Policy

- **Bug Reporting:** A dedicated support portal and email address will be provided for stakeholders to report bugs. All reports will be logged, acknowledged, and triaged according to the SLA.
- **Release Cycle:** Patches for critical bugs will be deployed immediately. Major and minor bug fixes will be bundled and released in scheduled maintenance updates to minimize disruption.
- **Operating System (OS) Upgrade Policy:** When a new major version of Android or iOS is released to the public, the mobile applications will be proactively tested for compatibility. A

compatibility update will be developed and rolled out within 45-60 days of the official OS release to ensure uninterrupted service for all users.

10.1. Detailed Cost Breakdown per Deliverable

The total project cost is broken down by the major deliverables to ensure transparency.

Deliverable	Description
Citizen Mobile App (Android & iOS)	Design, development, and testing of the citizen-facing mobile application.
Officer Mobile App (Android & iOS)	Design, development, and testing of the internal officer mobile application.
Admin & Citizen Web Consoles	Development of the two web-based management and registration portals.
Backend APIs, Integrations & Database	Development of all server-side logic, databases, and third-party integrations.
Data Migration	Middleware development and execution of the data migration plan from the existing portal.
Total Development Cost	
Maintenance & Support (First 6 Months)	Standard 6-month support plan for all components post-Go-Live.

10.2. External & Third-Party Costs

The following ongoing operational costs are external to this proposal and are to be borne directly by the client. These are variable costs that depend on usage and the specific vendor chosen.

- **Infrastructure (Hosting & Cloud):** Secure hosting on NIC / MeghRaj Cloud. This includes costs for virtual machines, storage, bandwidth, and load balancers.
- **Google Maps API:** Billed by Google based on the volume of map loads, direction requests, and other API calls.
- **SMS/OTP Gateway:** Charged on a per-SMS basis by the chosen gateway provider (e.g., MSG91, Twilio).

- **CERT-In Security Audit:** A one-time cost paid directly to the CERT-In empanelled vendor for the mandatory VAPT audit required before Go-Live.

10.3. Payment Milestones

Payments for the total development cost will be structured according to the following project milestones:

Milestone	Percentage of Total Development Cost
SRS and Design Approval	15%
Completion of Backend & Admin Console Development, Mobile App Development & Integration with backend.	60%
Successful Migration,UAT Sign-off	15%
Security Audit and Go-Live	10%

10.4. Terms and Conditions

- All costs quoted are exclusive of applicable taxes (e.g., GST).
- Any change in scope or requirements outside of the deliverables outlined in this proposal will be subject to a formal Change Request process and may incur additional costs and timeline adjustments.
- Payment is due within 15 days of the date of invoice receipt for each milestone.
- This proposal is valid for a period of 30 days from the date of submission.

11. Assumptions & Prerequisites

The successful and timely execution of this project is based on the following assumptions:

- **Stakeholder Availability:** Key stakeholders and subject matter experts from the Delhi Police will be available for requirement clarification, regular feedback sessions, and UAT participation as per the mutually agreed project schedule.
- **Third-Party Credentials & Accounts:** All necessary API keys, credentials, and billing accounts for third-party services (SMS Gateway, Google Maps API, App Store developer accounts) will be procured and provided by the client in a timely manner.
- **Existing Data Access:** Unhindered access to the existing senior citizen portal's database, documentation, and technical team will be provided for the data migration process.
- **Content & Approvals:** All necessary content (e.g., text for notifications, branding guidelines, official logos) and formal approvals for deployment and public release will be provided by the client when required.

- **Hosting Environment Setup:** The server infrastructure on NIC / MeghRaj Cloud will be provisioned and made available to the development team as per the project timeline.